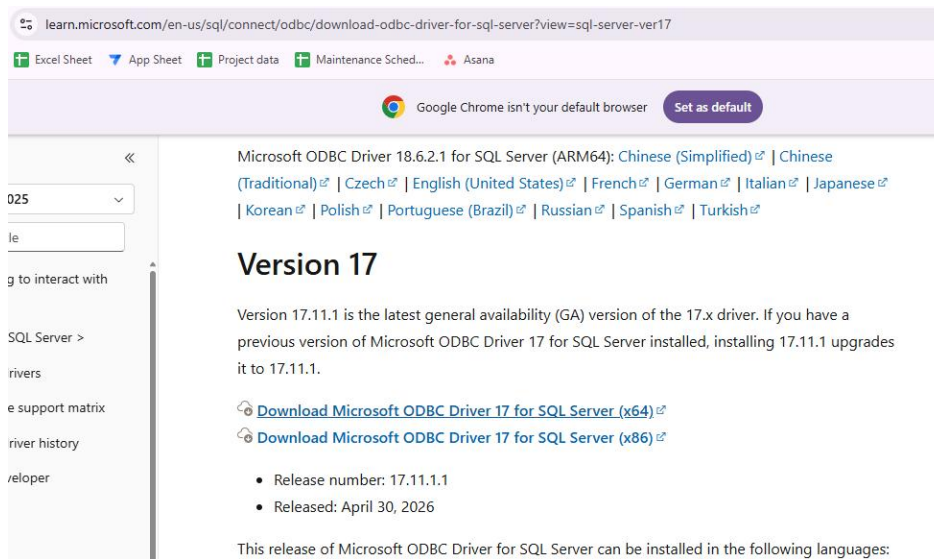


1. Install Python environment on your Windows

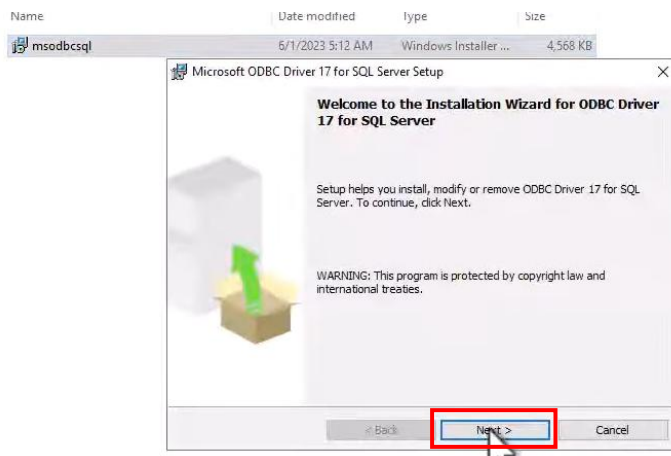
ODBC SQL Server Connection

2. Install ODBC Driver17

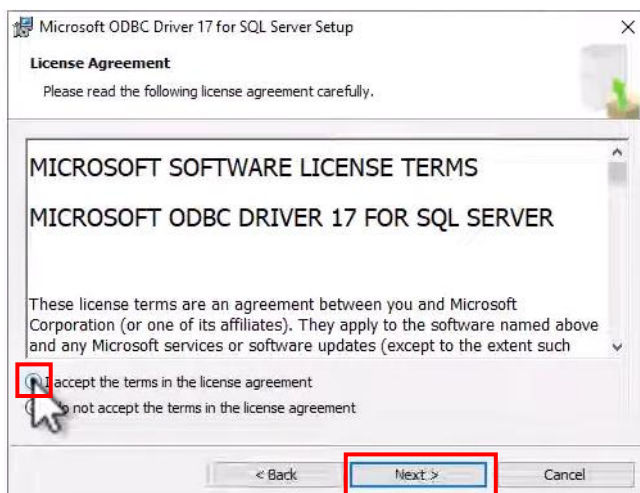
Url: <https://learn.microsoft.com/en-us/sql/connect/odbc/download-odbc-driver-for-sql-server?view=sql-server-ver17>



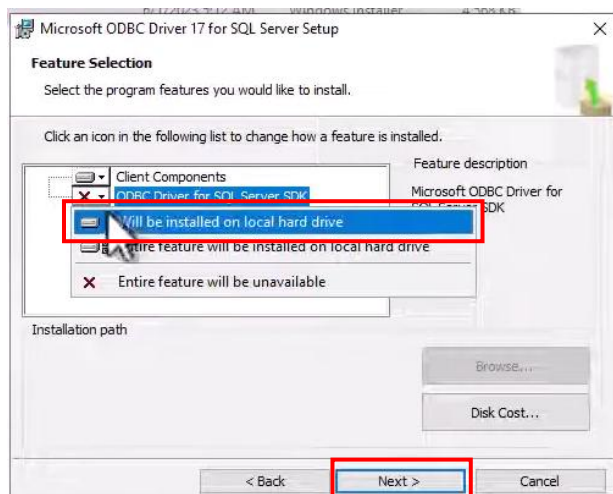
Click next



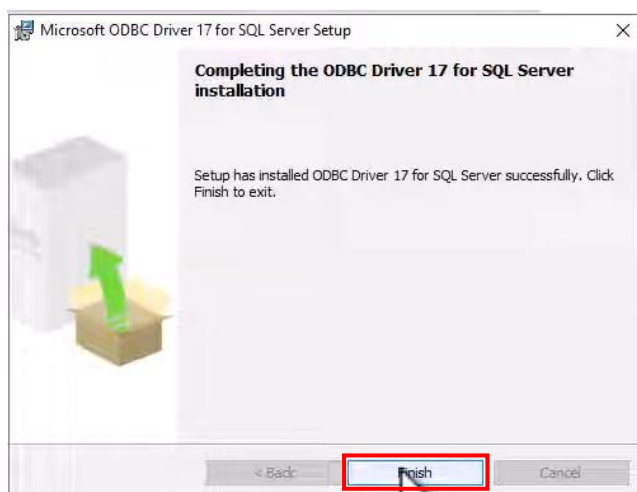
Check I accept the term in the license agreement > Next



Click ODBC Driver for SQL Server SDK > Will be installed on local hard drive > Next

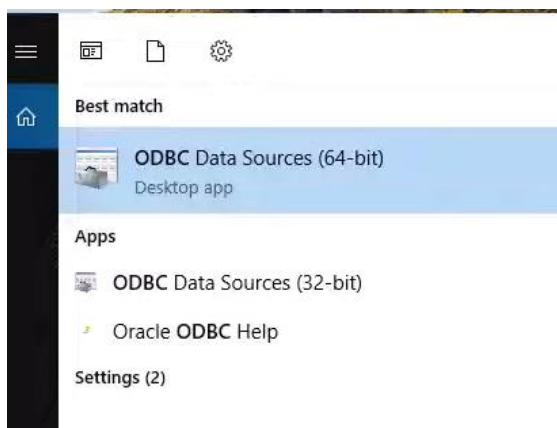


Next > Finish

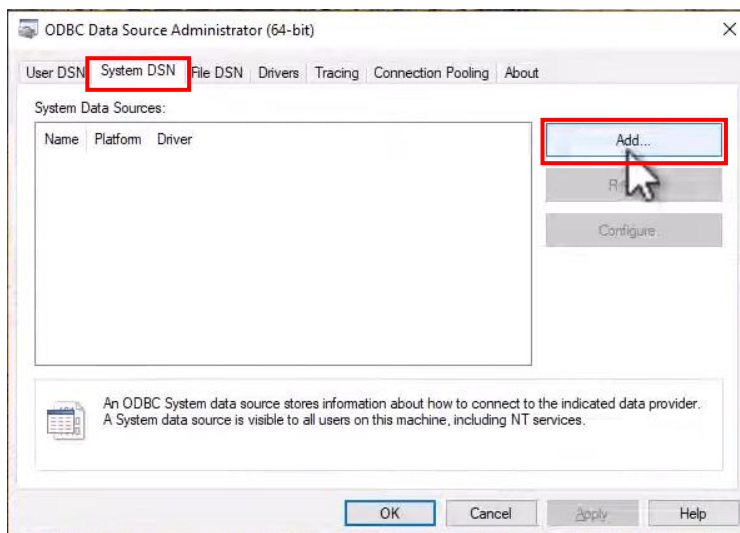


Configure ODBC

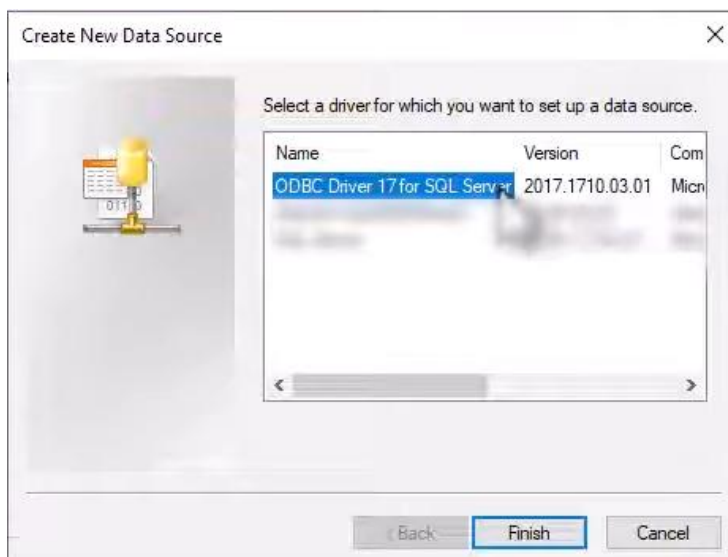
3. Search ODBC Data Sources (64-bit)



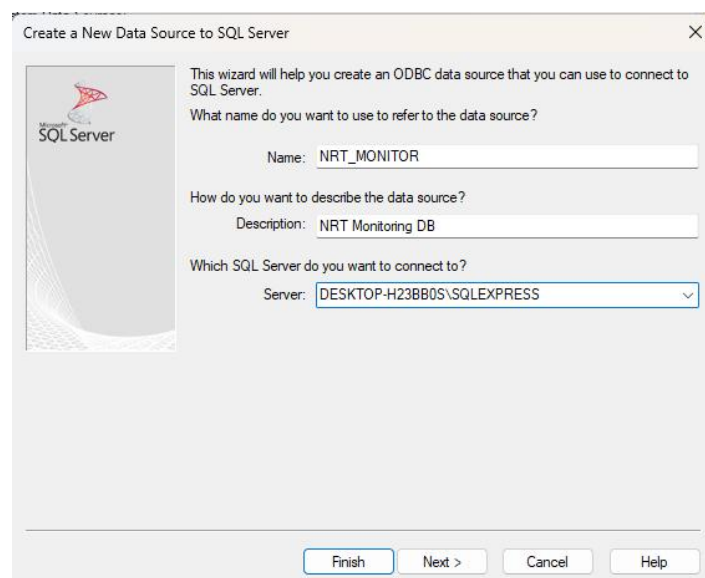
Click System DSN > Add



Choose ODBC Driver 17 for SQL Server

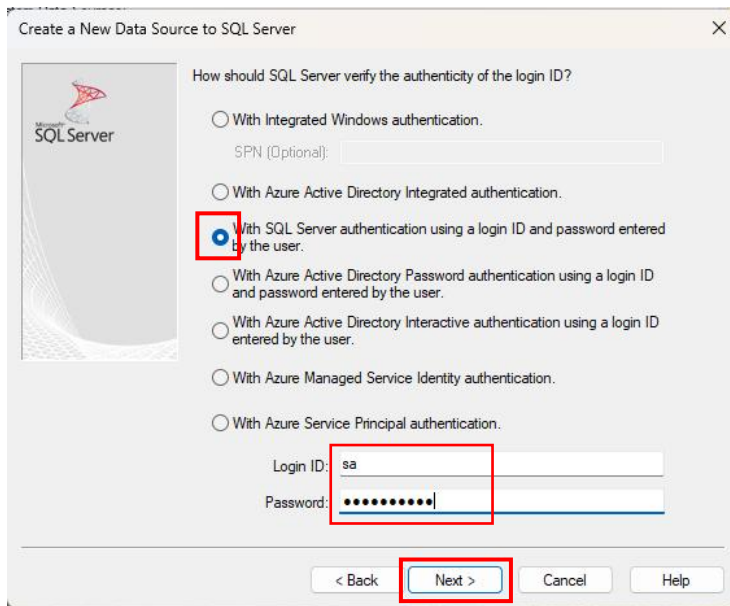


Fill in the details accordingly



Name: NRT_MONITOR
Description: NRT Monitoring DB
Server: <Choose your ssms server name>

If your server is authenticated check: “With SQL Server authentication using a login ID and password entered by the user”
Enter your login ID and password > Next



Create a New Data Source to SQL Server

How should SQL Server verify the authenticity of the login ID?

- ☐ With Integrated Windows authentication.
SPN (Optional):
- ☐ With Azure Active Directory Integrated authentication.
- ☒ With SQL Server authentication using a login ID and password entered by the user.
- ☐ With Azure Active Directory Password authentication using a login ID and password entered by the user.
- ☐ With Azure Active Directory Interactive authentication using a login ID entered by the user.
- ☐ With Azure Managed Service Identity authentication.
- ☐ With Azure Service Principal authentication.

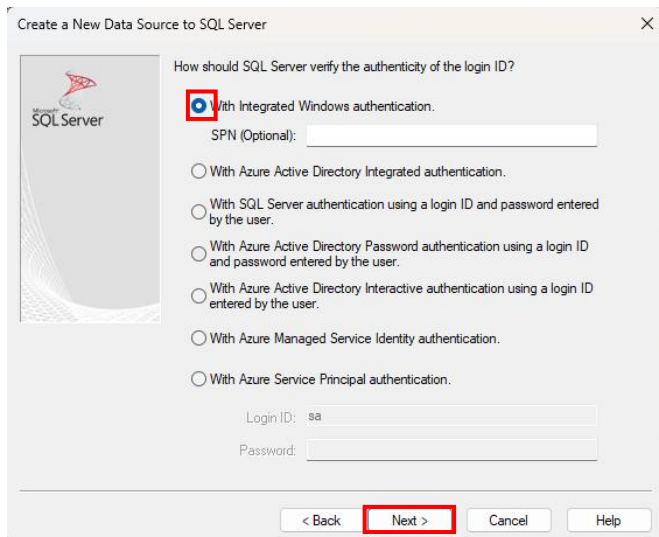
Login ID: sa

Password:

< Back Next > Cancel Help

Login ID: sa
Password: qwerty7890

If your server is unauthenticated, choose “With Integrated Windows authentication” > Next



Create a New Data Source to SQL Server

How should SQL Server verify the authenticity of the login ID?

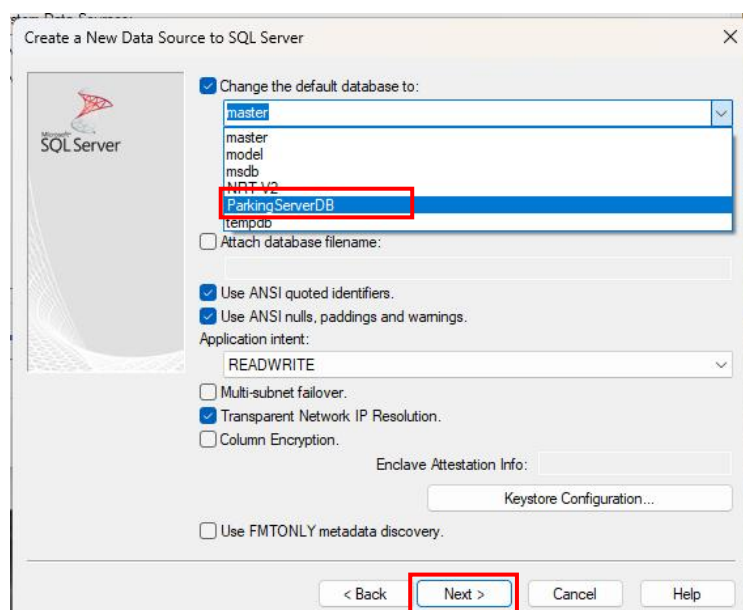
- ☒ With Integrated Windows authentication.
SPN (Optional):
- ☐ With Azure Active Directory Integrated authentication.
- ☐ With SQL Server authentication using a login ID and password entered by the user.
- ☐ With Azure Active Directory Password authentication using a login ID and password entered by the user.
- ☐ With Azure Active Directory Interactive authentication using a login ID entered by the user.
- ☐ With Azure Managed Service Identity authentication.
- ☐ With Azure Service Principal authentication.

Login ID: sa

Password:

< Back Next > Cancel Help

Change the default database to: ParkingServerDB > Next



Create a New Data Source to SQL Server

☒ Change the default database to:

master
master
model
msdb
NTPT_V2
ParkingServerDB
tempdb

☐ Attach database filename:

☒ Use ANSI quoted identifiers.
☒ Use ANSI nulls, paddings and warnings.

Application intent:
READWRITE

☐ Multi-subnet failover.
☒ Transparent Network IP Resolution.
☐ Column Encryption.

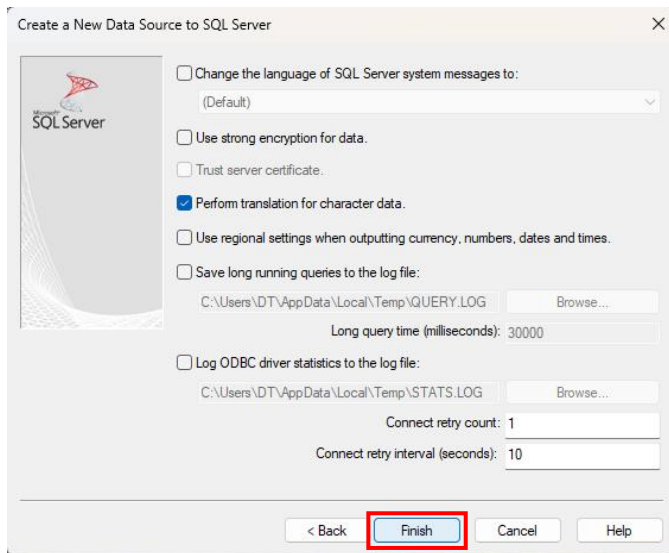
Enclave Attestation Info:

Keystore Configuration...

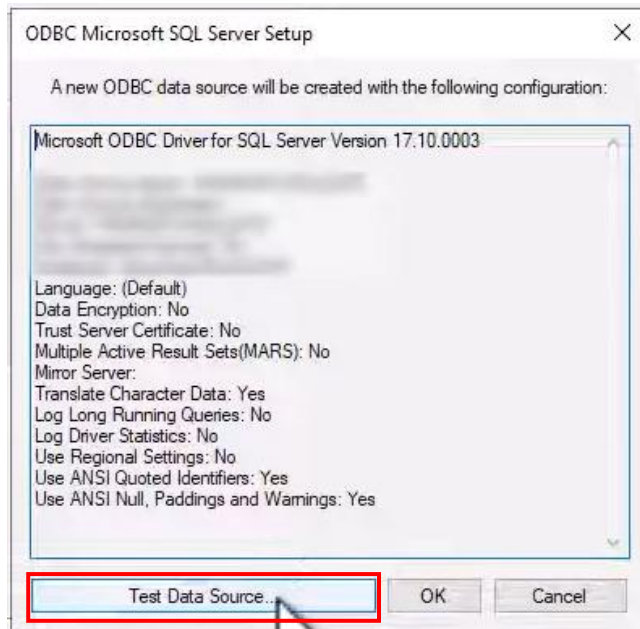
☐ Use FMTONLY metadata discovery.

< Back Next > Cancel Help

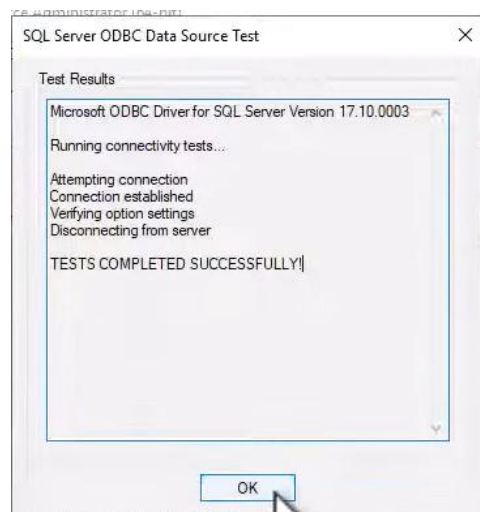
Click Finish



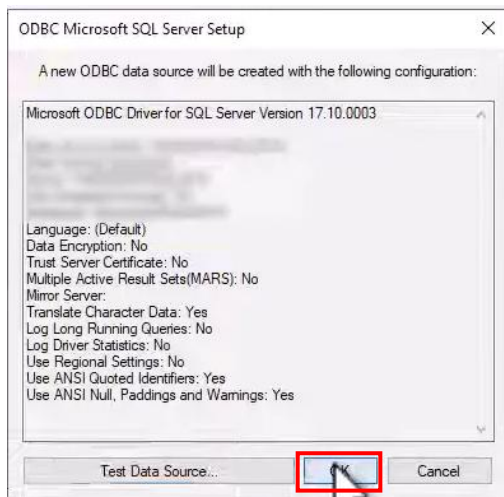
Click Test Data Source



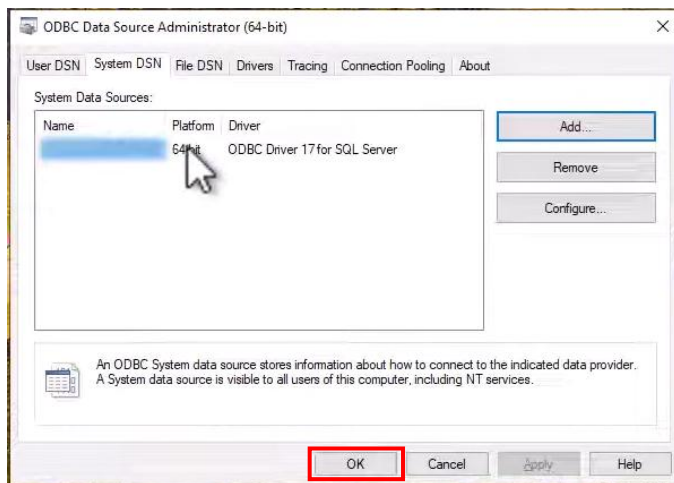
If you see "Connection established" means connection success



Click OK



Click OK



Run Instruction

1. ****Change directory****

```
cd NRT_Monitor
```

2. ****Create a virtual environment****

```
py -m venv venv  
.\venv\Scripts\activate # On Linux Mac OS: source venv/bin/activate
```

- If errors occurred, run this:

```
Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser
```

3. ****Install dependencies****

```
pip install -r requirements.txt
```

4. **Credentials configuration**

In the Agent.py

```
SQL_SERVER = r"your_server_name" # eg.,: "(local)\SQLEXPRESS" or "DESKTOP-TAICUPD\SQLEXPRESS"  
DATABASE = "NRT-V2"  
SITE_NAME = "example: Penang Courtyard"  
SITE_CODE = DATABASE  
API_URL = "the_central_server_api_url_or_local" # eg.,: "http://192.168.1.43:5000/api/nrt-status"  
# RDP central server:  
# API_URL = "http://192.168.1.180:5000/api/nrt-status"  
ALERT_THRESHOLD_HOURS = 48  
COLLECTION_INTERVAL_MINUTES = 40  
  
CONNECTION_STRING = (  
    "DRIVER={ODBC Driver 17 for SQL Server};"  
    f"SERVER={SQL_SERVER};"  
    f"DATABASE={DATABASE};"  
    # "UID=your_ssms_uid;" # sa  
    # "PWD=your_ssms_pw;" # qwerty7890  
    "Trusted_Connection=yes;" # Comment this if using SQL auth and provide UID/PWD  
)
```

5. ****Run the app****

Place both program one in the Central Server, another in the client site

On the Central Server, run:

```
python app.py
```

On the client side, run:

```
python agent.py
```

Output:

127.0.0.1:5000/?site=test-site-nrt-v2-local-sqlexpress#site-detail

Sites below currently have failed NRT detection based on TB_TNG_CONTROL_BATCH.

Test Site

5 control batch rows

SITE DETAIL

Test Site (NRT-V2)

(local)\SQLEXPRESS | NRT-V2

NRT alert - 5 control batch records pending, oldest operdate is 120 hours old

TERMINAL COUNT

1

BATCH COUNT

5

LATEST OPERDATE

2026-05-07 15:29:46

OLDEST OPERDATE

2026-05-03 15:29:46

OLDEST AGE HOURS

120

RECEIVED AT

2026-05-08 15:48:00

Red

TB_TNG_CONTROL_BATCH

5 rows

Site ID	Terminal ID	Batch No.	Operdate	EOJ Date Time	Record Count	Age Hours	Status
D67	P01	105	2026-05-07 15:29:46	2026-05-08 15:29:46	20	24	
D67	P01	104	2026-05-06 15:29:46	2026-05-07 15:29:46	15	48	
D67	P01	103	2026-05-05 15:29:46	2026-05-06 15:29:46	8	72	
D67	P01	102	2026-05-04 15:29:46	2026-05-05 15:29:46	12	96	
D67	P01	101	2026-05-03 15:29:46	2026-05-04 15:29:46	10	120	